

# OnStepX Command Reference

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This reference is derived from the current command handlers in the source tree, primarily the `*.command.cpp` files plus the shared command processor in `src/libApp/commands/ProcessCmds.cpp`.

Scope notes:

- This covers the command families implemented in the current tree, not older wiki pages.
- Many commands are compile-time feature gated. If a subsystem is not enabled in the build, its commands will not exist.
- Plugin-defined commands are not documented here because they come from the active plugin set, not the core command handlers.
- A few command comments in source are stale. Where the code and comment disagree, this document follows the code path.

## Protocol Basics

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### Frames

- Normal ASCII command frame: `:CC...#`
- Checksummed frame: `;CC...CCS#`
  - `;` starts a checksummed command.
  - The command processor validates the checksum and sequence character before dispatch.
- The parser ignores ASCII space, LF, and CR.
- Internal command buffer size is 80 bytes including framing, so current implementations are not limited to the older 40-character LX200 convention.

### Command Parsing

- `CC` is effectively a 1- or 2-character command family. The rest of the payload is the parameter string.
- Replies are either:
  - payload replies, usually terminated with `#`

- numeric/boolean replies, returned as 1 for success and 0 for failure
- Some commands suppress the trailing # on purpose.

## Global Reply/Error Behavior

- Commands that leave `numericReply=true` return:
  - 1 for `CE_NONE` or `CE_1`
  - 0 for any other error
- `:GE#` returns the last command error code as a 2-digit payload.

## Command Error Codes

These are the codes returned by `:GE#` :

| Code | Symbol                                | Meaning                             |
|------|---------------------------------------|-------------------------------------|
| 00   | <code>CE_NONE</code>                  | No error                            |
| 01   | <code>CE_0</code>                     | False/fail without a protocol error |
| 02   | <code>CE_CMD_UNKNOWN</code>           | Unknown command                     |
| 03   | <code>CE_REPLY_UNKNOWN</code>         | Invalid reply                       |
| 04   | <code>CE_PARAM_RANGE</code>           | Parameter out of range              |
| 05   | <code>CE_PARAM_FORM</code>            | Bad parameter format                |
| 06   | <code>CE_ALIGN_FAIL</code>            | Align failed                        |
| 07   | <code>CE_ALIGN_NOT_ACTIVE</code>      | Align not active                    |
| 08   | <code>CE_NOT_PARKED_OR_AT_HOME</code> | Not parked or at home               |
| 09   | <code>CE_PARKED</code>                | Already parked                      |
| 10   | <code>CE_PARK_FAILED</code>           | Park failed                         |
| 11   | <code>CE_NOT_PARKED</code>            | Not parked                          |
| 12   | <code>CE_NO_PARK_POSITION_SET</code>  | No park position set                |
| 13   | <code>CE_GOTO_FAIL</code>             | Goto failed                         |
| 14   | <code>CE_LIBRARY_FULL</code>          | Library full                        |

|    |                            |                             |
|----|----------------------------|-----------------------------|
| 15 | CE_SLEW_ERR_BELOW_HORIZON  | Target below horizon limit  |
| 16 | CE_SLEW_ERR_ABOVE_OVERHEAD | Target above overhead limit |
| 17 | CE_SLEW_ERR_IN_STANDBY     | Controller in standby       |
| 18 | CE_SLEW_ERR_IN_PARK        | Mount parked                |
| 19 | CE_SLEW_IN_SLEW            | Goto already active         |
| 20 | CE_SLEW_ERR_OUTSIDE_LIMITS | Outside configured limits   |
| 21 | CE_SLEW_ERR_HARDWARE_FAULT | Hardware fault              |
| 22 | CE_SLEW_IN_MOTION          | Mount already in motion     |
| 23 | CE_SLEW_ERR_UNSPECIFIED    | Other slew error            |
| 25 | CE_1                       | Explicit true/success       |

## Global Processor Commands

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These come from `ProcessCmds.cpp` , not from a subsystem `*.command.cpp` .

| Command | Reply                          | Description          |
|---------|--------------------------------|----------------------|
| :SBB#   | 1 then switches to 460800 baud | Set serial baud rate |
| :SBA#   | 1 then switches to 230400 baud | Set serial baud rate |
| :SB0#   | 1 then switches to 115200 baud | Set serial baud rate |
| :SB1#   | 1 then switches to 56700 baud  | Set serial baud rate |
| :SB2#   | 1 then switches to 38400 baud  | Set serial baud rate |
| :SB3#   | 1 then switches to 28800 baud  | Set serial baud rate |
|         |                                |                      |

|        |                                   |                                                                               |
|--------|-----------------------------------|-------------------------------------------------------------------------------|
| :SB4#  | 1 then switches to 19200 baud     | Set serial baud rate                                                          |
| :SB5#  | 1 then switches to 14400 baud     | Set serial baud rate                                                          |
| :SB6#  | 1 then switches to 9600 baud      | Set serial baud rate                                                          |
| :SB7#  | 1 then switches to 4800 baud      | Set serial baud rate                                                          |
| :SB8#  | 1 then switches to 2400 baud      | Set serial baud rate                                                          |
| :SB9#  | 1 then switches to 1200 baud      | Set serial baud rate                                                          |
| :GE#   | CC#                               | Get last command error code                                                   |
| <0x06> | A , P , or CK_FAIL style response | Special LX200 status query/checksum-fail path handled by the buffer/processor |

## Dispatcher Notes

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In the full telescope build, command dispatch order is:

1. mount
2. guide
3. GPIO/plugin status helpers
4. goto
5. park
6. library
7. site
8. limits
9. home
10. PEC
11. axis 1 / axis 2

12. rotator
13. focuser
14. auxiliary features

That matters because some command names are reused by different subsystems, especially `:hP#` and `:hR#`. In an integrated mount build, mount park/unpark wins before the rotator/focuser handlers ever see those frames. The rotator/focuser `h*` commands are still valid for standalone or remote-node builds.

## Core Telescope / Firmware / Environment

| Command                 | Reply                           | Description                                                                                           |
|-------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------|
| <code>:B+#</code>       | none                            | Increase reticle brightness                                                                           |
| <code>:B-#</code>       | none                            | Decrease reticle brightness                                                                           |
| <code>:ECtext#</code>   | none                            | Echo text to debug output. <code>_</code> becomes space, trailing <code>&amp;</code> becomes newline. |
| <code>:ERESet#</code>   | none                            | Reset the MCU                                                                                         |
| <code>:ENVRESET#</code> | text                            | Mark NV storage to be cleared on next boot                                                            |
| <code>:ESPFLASH#</code> | 1 /none depending on build path | Put addon/ESP device into firmware flashing mode when supported                                       |
| <code>:GVD#</code>      | MTH DD YYYY#                    | Firmware build date                                                                                   |
| <code>:GVM#</code>      | name version#                   | Firmware name plus version                                                                            |
| <code>:GVN#</code>      | M.mm...#                        | Firmware version number                                                                               |
| <code>:GVP#</code>      | name#                           | Product name                                                                                          |
| <code>:GVT#</code>      | HH:MM:SS#                       | Firmware build time                                                                                   |
| <code>:GVC#</code>      | string#                         | Firmware config/product description                                                                   |
| <code>:GVH#</code>      | string#                         | Firmware hardware/pinmap string                                                                       |
| <code>:GX9A#</code>     | sn.n#                           | Ambient temperature in C                                                                              |
| <code>:GX9B#</code>     | n.n#                            | Pressure in mbar                                                                                      |

|             |         |                              |
|-------------|---------|------------------------------|
| :GX9C#      | n.n#    | Relative humidity in percent |
| :GX9E#      | sn.n#   | Dew point in C               |
| :GX9F#      | n# or 0 | MCU temperature in C         |
| :SX9A,sn.n# | 0/1     | Set ambient temperature      |
| :SX9B,n.n#  | 0/1     | Set pressure                 |
| :SX9C,n.n#  | 0/1     | Set humidity                 |

## Time / Date / Site

| Command | Reply           | Description                                |
|---------|-----------------|--------------------------------------------|
| :Ga#    | HH:MM:SS#       | Local standard time, 12-hour format        |
| :GC#    | MM/DD/YY#       | Local standard calendar date               |
| :Gc#    | 24#             | Local time format indicator                |
| :GG#    | sHH:MM#         | UTC offset to add to local time to get UT1 |
| :Gg#    | sDDD*MM#        | Current longitude                          |
| :GgH#   | sDDD*MM:SS.SSS# | Longitude, highest precision               |
| :GL#    | HH:MM:SS#       | Local standard time, 24-hour format        |
| :GLH#   | HH:MM:SS.SSSS#  | Local standard time, highest precision     |
| :GM#    | string#         | Site 1 name                                |
| :GN#    | string#         | Site 2 name                                |
| :G0#    | string#         | Site 3 name                                |
| :GP#    | string#         | Site 4 name                                |
| :GS#    | HH:MM:SS#       | Sidereal time                              |
| :GSH#   | HH:MM:SS.ss#    | Sidereal time, highest precision           |
| :Gt#    | sDD*MM#         | Current latitude                           |

|                      |                |                                                                  |
|----------------------|----------------|------------------------------------------------------------------|
| :GtH#                | sDD*MM:SS.SSS# | Latitude, highest precision                                      |
| :Gv#                 | sn.n#          | Elevation in meters                                              |
| :GX80#               | HH:MM:SS.ss#   | UT1 time                                                         |
| :GX81#               | MM/DD/YY#      | UT1 date                                                         |
| :GX89#               | 0 or 1         | Date/time ready status. 0 means ready, 1 means not ready.        |
| :SCMM/DD/YY#         | 0/1            | Set local date                                                   |
| :SCMM/DD/YYYY#       | 0/1            | Set local date                                                   |
| :SGsHH#              | 0/1            | Set UTC offset                                                   |
| :SGsHH:MM#           | 0/1            | Set UTC offset. Comments indicate MM should be 00 , 30 , or 45 . |
| :Sg(s)DDD*MM#        | 0/1            | Set longitude                                                    |
| :Sg(s)DDD*MM:SS#     | 0/1            | Set longitude                                                    |
| :Sg(s)DDD*MM:SS.SSS# | 0/1            | Set longitude                                                    |
| :SLHH:MM:SS#         | 0/1            | Set local time                                                   |
| :SLHH:MM:SS.SSS#     | 0/1            | Set local time                                                   |
| :SMname#             | 0/1            | Set site 1 name, max 15 chars                                    |
| :SNname#             | 0/1            | Set site 2 name, max 15 chars                                    |
| :S0name#             | 0/1            | Set site 3 name, max 15 chars                                    |
| :SPname#             | 0/1            | Set site 4 name, max 15 chars                                    |
| :StsDD*MM#           | 0/1            | Set latitude                                                     |
| :StsDD*MM:SS#        | 0/1            | Set latitude                                                     |
| :StsDD*MM:SS.SSS#    | 0/1            | Set latitude                                                     |
| :SUs.s#              | 0/1            | Set DUT1 correction in seconds, nominally -0.9 to +0.9           |

|              |      |                         |
|--------------|------|-------------------------|
| :Svsn.n#     | 0/1  | Set elevation in meters |
| :W0# .. :W3# | none | Select active site slot |
| :W?#         | n#   | Query active site slot  |

## Mount Position / Tracking / Rates

Available when MOUNT\_PRESENT is enabled.

| Command | Reply                    | Description                                              |
|---------|--------------------------|----------------------------------------------------------|
| :GA#    | sDD*MM# or<br>sDD*MM'SS# | Current mount altitude                                   |
| :GAH#   | sDD*MM'SS.SSS#           | Current mount altitude, highest precision                |
| :GD#    | sDD*MM# or<br>sDD*MM:SS# | Current declination                                      |
| :GDH#   | sDD*MM:SS.SSS#           | Current declination, highest precision                   |
| :GR#    | HH:MM.T# or<br>HH:MM:SS# | Current right ascension                                  |
| :GRH#   | HH:MM:SS.SSSS#           | Current right ascension, highest precision               |
| :GT#    | n.nnnnn# or<br>0#        | Tracking rate in Hz, 0 if not tracking                   |
| :GZ#    | DDD*MM# or<br>DDD*MM'SS# | Current azimuth                                          |
| :GZH#   | DDD*MM'SS.SSS#           | Current azimuth, highest precision                       |
| :GXTD#  | n.nnnnnnnn#              | Tracking rate offset, Dec, arcsec per<br>sidereal second |
| :GXTR#  | n.nnnnnnnn#              | Tracking rate offset, RA, arcsec per<br>sidereal second  |
| :GX40#  | DDD*MM:SS#               | Axis1 instrument angle                                   |
| :GX41#  | DDD*MM:SS#               | Axis2 instrument angle                                   |



|          |               |                                                                                             |
|----------|---------------|---------------------------------------------------------------------------------------------|
| :GX42#   | n.nnnnnn#     | Axis1 instrument angle in decimal degrees                                                   |
| :GX43#   | n.nnnnnn#     | Axis2 instrument angle in decimal degrees                                                   |
| :GX44#   | n#            | Axis1 encoder count                                                                         |
| :GX45#   | n#            | Axis2 encoder count                                                                         |
| :GXE4#   | n#            | Axis1 steps per degree                                                                      |
| :GXE5#   | n#            | Axis2 steps per degree                                                                      |
| :GXEE#   | 0# , 1# , ... | Mount coordinate mode ( MOUNT_COORDS - 1 )                                                  |
| :GXEF#   | 1 or 0        | Axis2 tangent-arm presence flag                                                             |
| :GXEG#   | 1 or 0        | Axis1 sector-gear presence flag                                                             |
| :GXEM#   | n#            | Current mount type                                                                          |
| :GXF3#   | sn.nnnnnn#    | Axis1 step frequency                                                                        |
| :GXF4#   | sn.nnnnnn#    | Axis2 step frequency                                                                        |
| :GXFA#   | 50%#          | Reported workload placeholder                                                               |
| :GXFF#   | n.nnnnnn#     | Axis1 index position                                                                        |
| :GXFG#   | n.nnnnnn#     | Axis2 index position                                                                        |
| :STn.n#  | 0/1           | Set tracking rate in Hz. 0 stops tracking.                                                  |
| :SE0#    | 0/1           | Save absolute encoder origin, or initialize mount coordinate memory at home, when supported |
| :SX40,n# | 0/1           | Stage encoder axis1 angle in degrees                                                        |
| :SX41,n# | 0/1           | Stage encoder axis2 angle in degrees                                                        |
| :SX42,1# | 0/1           | Sync mount from staged encoder axis values                                                  |
| :SX43,0# | 0/1           | Allow SWS to control sync mode                                                              |
|          |               |                                                                                             |

|                     |      |                                                                                                  |
|---------------------|------|--------------------------------------------------------------------------------------------------|
| :SX44,deg1,deg2[a]# | 0/1  | Stage and sync both encoder axes, append a when both SWS encoder values are absolute and trusted |
| :SXEM,n#            | 0/1  | Set mount type for next restart                                                                  |
| :SXTD,n.n#          | 0/1  | Set Dec tracking rate offset, arcsec per sidereal second                                         |
| :SXTR,n.n#          | 0/1  | Set RA tracking rate offset, arcsec per sidereal second                                          |
| :TS#                | none | Solar tracking rate                                                                              |
| :TK#                | none | King tracking rate                                                                               |
| :TL#                | none | Lunar tracking rate                                                                              |
| :TQ#                | none | Sidereal tracking rate                                                                           |
| :T+#                | none | Increase master sidereal clock by 0.02 Hz                                                        |
| :T-#                | none | Decrease master sidereal clock by 0.02 Hz                                                        |
| :TR#                | none | Reset master sidereal clock                                                                      |
| :Te#                | 0/1  | Enable tracking                                                                                  |
| :Td#                | 0/1  | Disable tracking                                                                                 |
| :To#                | 0/1  | Enable full compensation model                                                                   |
| :Tr#                | 0/1  | Enable refraction compensation                                                                   |
| :Tn#                | 0/1  | Disable compensation                                                                             |
| :T1#                | 0/1  | Single-axis tracking mode                                                                        |
| :T2#                | 0/1  | Dual-axis tracking mode                                                                          |
| :\$BDn#             | 0/1  | Set Dec/Alt backlash in arcsec                                                                   |
| :\$BRn#             | 0/1  | Set RA/Azm backlash in arcsec                                                                    |
| :%BD#               | n#   | Get Dec/Alt backlash in arcsec                                                                   |

|       |    |                               |
|-------|----|-------------------------------|
| :%BR# | n# | Get RA/Azm backlash in arcsec |
|-------|----|-------------------------------|

## Mount Type Values

Current core constants are:

| Value | Mount type |
|-------|------------|
| 1     | GEM        |
| 2     | FORK       |
| 3     | ALTAZM     |
| 4     | ALTALT     |
| 5     | GEM_TA     |
| 6     | GEM_TAC    |
| 7     | FORK_TA    |
| 8     | FORK_TAC   |
| 9     | ALTAZM_UNL |

## Goto / Sync / Alignment

| Command      | Reply              | Description                                                   |
|--------------|--------------------|---------------------------------------------------------------|
| :AW#         | 1                  | Write alignment model to NV                                   |
| :A?#         | mno#               | Alignment status: max stars, current star, last required star |
| :A1# .. :A9# | 0/1                | Start manual alignment with that many stars                   |
| :A+#         | 0/1                | Accept current align point                                    |
| :CS#         | none               | Sync to current target coordinates                            |
| :CM#         | N/A# or E1# .. E9# | Sync to current catalog/database object                       |

|                   |                                        |                                                |
|-------------------|----------------------------------------|------------------------------------------------|
| :D#               | 0x7f# while moving,<br>otherwise raw # | LX200 distance-bar style motion<br>indicator   |
| :Gr#              | HH:MM.T# or<br>HH:MM:SS#               | Get target RA                                  |
| :GrH#             | HH:MM:SS.SSSS#                         | Get target RA, highest precision               |
| :Gd#              | sDD*MM# or<br>sDD*MM:SS#               | Get target Dec                                 |
| :GdH#             | sDD*MM:SS.SSS#                         | Get target Dec, highest precision              |
| :Gal#             | sDD*MM# or<br>sDD*MM'SS#               | Get target altitude                            |
| :GaH#             | sDD*MM'SS.SSS#                         | Get target altitude, highest precision         |
| :Gz#              | DDD*MM# or<br>DDD*MM'SS#               | Get target azimuth                             |
| :GzH#             | DDD*MM'SS.SSS#                         | Get target azimuth, highest precision          |
| :MA#              | 0 .. 9                                 | Goto target Alt/Az                             |
| :MD#              | 0 , 1 , or 2                           | Destination pier side for current<br>target    |
| :MN#              | 0 .. 9                                 | Goto current position on opposite<br>pier side |
| :MNe#             | 0 .. 9                                 | Force same-position goto to east<br>pier side  |
| :MNw#             | 0 .. 9                                 | Force same-position goto to west<br>pier side  |
| :MP#              | 0 .. 9                                 | Polar-align goto                               |
| :MS#              | 0 .. 9                                 | Goto current target                            |
| :SrHH:MM.T#       | 0/1                                    | Set target RA                                  |
| :SrHH:MM:SS#      | 0/1                                    | Set target RA                                  |
| :SrHH:MM:SS.SSSS# | 0/1                                    | Set target RA                                  |

|                   |     |                     |
|-------------------|-----|---------------------|
| :SdsDD*MM#        | 0/1 | Set target Dec      |
| :SdsDD*MM:SS#     | 0/1 | Set target Dec      |
| :SdsDD*MM:SS.SSS# | 0/1 | Set target Dec      |
| :SasDD*MM#        | 0/1 | Set target altitude |
| :SasDD*MM'SS#     | 0/1 | Set target altitude |
| :SasDD*MM'SS.SSS# | 0/1 | Set target altitude |
| :SzDDD*MM#        | 0/1 | Set target azimuth  |
| :SzDDD*MM'SS#     | 0/1 | Set target azimuth  |
| :SzDDD*MM'SS.SSS# | 0/1 | Set target azimuth  |

### :MS# / :MA# / :MN# / :MP# Return Codes

| Value | Meaning                  |
|-------|--------------------------|
| 0     | Goto accepted            |
| 1     | Below horizon limit      |
| 2     | Above overhead limit     |
| 3     | Controller in standby    |
| 4     | Mount parked             |
| 5     | Goto already in progress |
| 6     | Outside limits           |
| 7     | Hardware fault           |
| 8     | Already in motion        |
| 9     | Unspecified error        |

## Alignment Model Extended Commands

ALIGN\_MAX\_NUM\_STARS > 1 builds expose:

| Command                 | Reply      | Description                                               |
|-------------------------|------------|-----------------------------------------------------------|
| :GX00#                  | n#         | ax1Cor in arcsec                                          |
| :GX01#                  | n#         | ax2Cor in arcsec                                          |
| :GX02#                  | n#         | altCor in arcsec                                          |
| :GX03#                  | n#         | azmCor in arcsec                                          |
| :GX04#                  | n#         | doCor in arcsec                                           |
| :GX05#                  | n#         | pdCor in arcsec                                           |
| :GX06#                  | n#         | ffCor for FORK/ALTAZM, else 0                             |
| :GX07#                  | n#         | dfCor for GEM-style mounts, else 0                        |
| :GX08#                  | n#         | tfCor in arcsec                                           |
| :GX0a#                  | n#         | hcp in degrees                                            |
| :GX0b#                  | n#         | hca in arcsec                                             |
| :GX0c#                  | n#         | dcp in degrees                                            |
| :GX0d#                  | n#         | dca in arcsec                                             |
| :GX09#                  | n#         | Number of uploaded stars, then resets internal star index |
| :GX0A#                  | HH:MM:SS#  | Uploaded star actual HA                                   |
| :GX0B#                  | sDD*MM:SS# | Uploaded star actual Dec                                  |
| :GX0C#                  | HH:MM:SS#  | Uploaded star mount HA                                    |
| :GX0D#                  | sDD*MM:SS# | Uploaded star mount Dec                                   |
| :GX0E#                  | n#         | Uploaded star pier side, then advances star index         |
| :SX00,n# ..<br>:SX0d,n# | 0/1        | Set alignment model coefficients listed above             |
|                         |            |                                                           |

|                  |     |                                           |
|------------------|-----|-------------------------------------------|
| :SX09,0#         | 0/1 | Reset alignment upload state              |
| :SX09,1#         | 0/1 | Build model from uploaded stars           |
| :SX09,2#         | 0/1 | Force model active                        |
| :SX0A,HH:MM:SS#  | 0/1 | Upload actual HA for current star         |
| :SX0B,sDD*MM:SS# | 0/1 | Upload actual Dec for current star        |
| :SX0C,HH:MM:SS#  | 0/1 | Upload mount HA for current star          |
| :SX0D,sDD*MM:SS# | 0/1 | Upload mount Dec for current star         |
| :SX0E,n#         | 0/1 | Upload pier side and advance to next star |

## Goto Extended Settings

| Command                 | Reply                | Description                                                 |
|-------------------------|----------------------|-------------------------------------------------------------|
| :GX92#                  | n.nnn#               | Current slew period in us/step                              |
| :GX93#                  | n.nnn#               | Base/default slew period in us/step                         |
| :GX94#                  | n plus<br>optional N | Current pier side: 0 none, 1 east, 2 west                   |
| :GX95#                  | 0 or 1               | Auto meridian flip enabled                                  |
| :GX96#                  | E , W , B , or<br>A  | Preferred pier side                                         |
| :GX97#                  | n.n#                 | Current step rate in deg/s                                  |
| :GX99#                  | n.nnn#               | Fastest allowed slew period in us/step                      |
| :SX92,n.nnn#            | 0/1                  | Set current slew period in us/step                          |
| :SX93,[1-5]#            | none                 | Slew preset: 5 =50%, 4 =66.7%, 3 =100%,<br>2 =150%, 1 =200% |
| :SX95,0# or<br>:SX95,1# | 0/1                  | Disable/enable automatic meridian flip                      |
| :SX96,E#                | 0/1                  | Preferred pier side east                                    |

|                         |     |                                                   |
|-------------------------|-----|---------------------------------------------------|
| :SX96,W#                | 0/1 | Preferred pier side west                          |
| :SX96,B#                | 0/1 | Preferred pier side best                          |
| :SX96,A#                | 0/1 | Preferred pier side automatic                     |
| :SX98,0# or<br>:SX98,1# | 0/1 | Disable/enable pause at home during meridian flip |
| :SX99,1#                | 0/1 | Continue after pause at home                      |

## Guide / Manual Motion

| Command        | Reply | Description                                                  |
|----------------|-------|--------------------------------------------------------------|
| :GX90#         | n.nn# | Pulse guide rate                                             |
| :Mgdn#         | none  | Pulse guide for n ms in direction d where d is w, e, n, or s |
| :MGdn#         | 0/1   | Same as :Mgdn#, numeric form                                 |
| :Mw#           | none  | Move west at current guide rate                              |
| :Me#           | none  | Move east at current guide rate                              |
| :Mn#           | none  | Move north at current guide rate                             |
| :Ms#           | none  | Move south at current guide rate                             |
| :Mp#           | none  | Spiral-search motion                                         |
| :Q#            | none  | Stop all slews, abort goto                                   |
| :Qe# /<br>:Qw# | none  | Stop east/west motion                                        |
| :Qn# /<br>:Qs# | none  | Stop north/south motion                                      |
| :RAn.n#        | none  | Set axis1 custom guide rate in deg/s                         |
| :REn.n#        | none  | Set axis2 custom guide rate in deg/s                         |
|                |       |                                                              |



|                 |      |                                          |
|-----------------|------|------------------------------------------|
| :RG#            | none | Guide rate preset 1x                     |
| :RC#            | none | Centering rate preset 8x                 |
| :RM#            | none | Find rate preset 20x                     |
| :RF#            | none | Fast rate preset 48x                     |
| :RS#            | none | Slew rate preset, half current goto rate |
| :R0# ..<br>:R9# | none | Numeric guide-rate preset                |

## Park / Home / Limits / Status

### Park

| Command | Reply | Description                  |
|---------|-------|------------------------------|
| :hP#    | 0/1   | Park mount                   |
| :hQ#    | 0/1   | Set current position as park |
| :hR#    | 0/1   | Unpark mount                 |

### Home

| Command          | Reply                             | Description                                                                             |
|------------------|-----------------------------------|-----------------------------------------------------------------------------------------|
| :h?#             | hasSense,axis1offset,axis2offset# | Home status. Source comment mentions auto-home, but current code returns 3 fields only. |
| :hA0# /<br>:hA1# | none                              | Disable/enable automatic home at boot                                                   |
| :hC#             | none                              | Move to home                                                                            |
| :hC1,R#          | none                              | Toggle axis1 home-sense reversal                                                        |
| :hC1,n#          | none                              | Set axis1 home offset in arcsec                                                         |

|         |      |                                         |
|---------|------|-----------------------------------------|
| :hC2,R# | none | Toggle axis2 home-sense reversal        |
| :hC2,n# | none | Set axis2 home offset in arcsec         |
| :hF#    | none | Reset mount at home/cold-start position |

## Limits

For a visual explanation of confusing GEM east/west pier-side limit geometry, see [GOTO\\_NOTES.md](#).

| Command  | Reply | Description                        |
|----------|-------|------------------------------------|
| :Gh#     | sDD*# | Horizon limit                      |
| :Go#     | DD*#  | Overhead limit                     |
| :GXE9#   | n#    | East meridian limit in minutes     |
| :GXEA#   | n#    | West meridian limit in minutes     |
| :GXEe#   | n#    | Axis1 minimum limit in degrees     |
| :GXEw#   | n#    | Axis1 maximum limit in degrees     |
| :GXEB#   | n#    | Axis1 maximum limit in hours       |
| :GXEC#   | n#    | Axis2 minimum limit in degrees     |
| :GXED#   | n#    | Axis2 maximum limit in degrees     |
| :ShsDD#  | 0/1   | Set lower altitude limit           |
| :SoDD#   | 0/1   | Set overhead altitude limit        |
| :SXE9,n# | 0/1   | Set east meridian limit in minutes |
| :SXEA,n# | 0/1   | Set west meridian limit in minutes |

## Status

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

| Command  | Reply           | Description                                   |
|----------|-----------------|-----------------------------------------------|
| :Gm#     | E# , W# , or N# | Meridian pier side                            |
| :GU#     | status string   | Human-readable packed status                  |
| :Gu#     | packed bytes    | Bit-packed status                             |
| :GW#     | 4-char status   | Mount type, tracking, parked/home, align-done |
| :SX97,0# | 0/1             | Disable buzzer                                |
| :SX97,1# | 0/1             | Enable buzzer                                 |
| :SX97,2# | 0/1             | Beep                                          |
| :SX97,3# | 0/1             | Alert tone                                    |
| :SX97,4# | 0/1             | Click                                         |

### :GU# Status Characters

The reply is an ordered string assembled from active conditions. Characters currently used by the code include:

| Char | Meaning                    |
|------|----------------------------|
| n    | Not tracking               |
| N    | No goto active             |
| p    | Not parked                 |
| I    | Parking in progress        |
| P    | Parked                     |
| F    | Park failed                |
| e    | Sync-to-encoders-only mode |
| H    | At home                    |
| h    | Homing                     |
| B    | Auto-home at boot          |
|      |                            |

|                   |                                                  |
|-------------------|--------------------------------------------------|
| S                 | PPS synced                                       |
| G                 | Pulse guide active                               |
| g                 | Guide active                                     |
| r                 | Refraction compensation enabled                  |
| s                 | Single-axis compensation mode                    |
| t                 | Full compensation model enabled                  |
| (                 | Lunar tracking rate selected                     |
| 0                 | Solar tracking rate selected                     |
| k                 | King tracking rate selected                      |
| w                 | Meridian flip paused at home                     |
| u                 | Pause-at-home enabled                            |
| z                 | Buzzer enabled                                   |
| a                 | Automatic meridian flip enabled                  |
| R                 | PEC data recorded                                |
| / , , , ~ , ; , ^ | PEC state                                        |
| E , K , A , L     | GEM, FORK, ALTAZM, ALTALT                        |
| o , T , W         | Pier side none, east, west                       |
| final digits      | pulse-guide rate, guide rate, general error code |

### : Gu# Packed Status Layout

Gu is the binary/pseudo-binary status form. The current implementation fills bytes as:

| Byte | Contents                                                       |
|------|----------------------------------------------------------------|
| 0    | tracking/goto/PPS/pulse-guide plus compensation mode           |
| 1    | tracking-rate selection plus sync-to-encoders and guide-active |
| 2    | home/homing/auto-home/home-pause/buzzer/auto-flip              |

|   |                                 |
|---|---------------------------------|
| 3 | mount type plus pier side       |
| 4 | PEC state and PEC-recorded flag |
| 5 | park state                      |
| 6 | pulse-guide selection           |
| 7 | guide-rate selection            |
| 8 | general limits/error code       |

## Object Library

| Command       | Reply             | Description                                      |
|---------------|-------------------|--------------------------------------------------|
| :LB#          | none              | Previous object matching current constraints     |
| :LCn#         | none              | Select record number n                           |
| :LI#          | name,type#        | Current object name and type                     |
| :LIG#         | none              | Load current object into goto target             |
| :LR#          | name,type,ra,dec# | Current object info, then advance to next record |
| :LWname,type# | 0/1               | Write current target RA/Dec to next free record  |
| :LN#          | none              | Next object matching current constraints         |
| :L\$#         | 1                 | Move to catalog name record                      |
| :LD#          | none              | Clear current record                             |
| :LL#          | none              | Clear current catalog                            |
| :L!#          | none              | Clear all catalogs                               |
| :L?#          | n#                | Free records across all catalogs                 |
| :Lon#         | 0/1               | Select catalog 0..14                             |

Object type codes written/read by library commands are:

UNK , OC , GC , PN , DN , SG , EG , IG , KNT , SNR , GAL , CN , STR , PLA , CMT , AST

## PEC

Available when PEC support is enabled for axis 1.

| Command  | Reply      | Description                                          |
|----------|------------|------------------------------------------------------|
| :GX91#   | n#         | PEC analog value                                     |
| :GXE6#   | n.nnnnnnn# | Steps per sidereal second                            |
| :GXE7#   | n#         | Worm rotation steps from NV                          |
| :GXE8#   | n#         | PEC buffer size in seconds                           |
| :SXE7,n# | 0/1        | Set worm rotation steps                              |
| :VH#     | nnnnn#     | PEC index sense position in sidereal seconds         |
| :VR#     | snnn,nnn#  | Current PEC segment correction plus segment index    |
| :VRn#    | snnn#      | PEC correction for segment n                         |
| :Vrn#    | x0x1...x9# | Ten-byte hex frame of PEC data starting at segment n |
| :VS#     | n.nnnnnnn# | Steps per sidereal second of worm rotation           |
| :VW#     | nnnnnnn#   | Worm rotation steps                                  |
| :WR+#    | 0/1        | Rotate PEC table forward one second                  |
| :WR-#    | 0/1        | Rotate PEC table backward one second                 |
| :WRn,sn# | none       | Write PEC correction for segment n                   |
| :\$QZ+#  | none       | Enable PEC playback                                  |
| :\$QZ-#  | none       | Disable PEC                                          |
| :\$QZ/#  | none       | Arm PEC recording                                    |

|         |                                                |                     |
|---------|------------------------------------------------|---------------------|
| :\$QZZ# | none                                           | Clear PEC buffer    |
| :\$QZ!# | none                                           | Save PEC data to NV |
| :\$QZ?# | I# , p# , P# , r# , R# ,<br>optionally with .# | PEC status          |

PEC status characters from :\$QZ?# :

| Char | Meaning                    |
|------|----------------------------|
| I    | Ignore/off                 |
| p    | Ready to play              |
| P    | Playing                    |
| r    | Ready to record            |
| R    | Recording                  |
| .    | Index detected this second |

## :\$GXUa# Driver Status Flags

Local ASCII replies are comma-separated flag mnemonics:

| Flag | Meaning                  |
|------|--------------------------|
| ST   | Standstill               |
| 0A   | Output A open load       |
| 0B   | Output B open load       |
| GA   | Output A short to ground |
| GB   | Output B short to ground |
| 0T   | Over-temperature         |
| PW   | Over-temperature warning |
| GF   | Driver fault             |

# Focuser

Available in focuser-enabled or remote-focuser builds.

Addressing rules:

- `:FA#` returns the currently selected focuser number.
- `:FA1# .. :FA6#` selects the active focuser.
- `:F...#` uses the active focuser.
- `:F1...# .. :F6...#` directs a command to a specific focuser immediately.

Unit rules for local ASCII focuser commands:

- Uppercase `B`, `D`, `G`, `I`, `M`, `R`, `S` use microns.
- Lowercase `b`, `d`, `g`, `i`, `m`, `r`, `s` use raw steps.
- Other focuser commands are unitless or fixed-unit as noted below.

| Command                     | Reply                                         | Description                                                                                           |
|-----------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <code>:FA#</code>           | <code>n</code>                                | Get active focuser number                                                                             |
| <code>:FA1# .. :FA6#</code> | <code>0/1</code>                              | Select active focuser                                                                                 |
| <code>:hP#</code>           | <code>0/1</code>                              | Park all focusers in standalone/remote focuser builds                                                 |
| <code>:hR#</code>           | <code>0/1</code>                              | Unpark all focusers in standalone/remote focuser builds                                               |
| <code>:Fa#</code>           | <code>1</code>                                | Primary focuser present/selected                                                                      |
| <code>:FT#</code>           | <code>M1#</code> ,<br><code>S3#</code> , etc. | Focuser status plus goto-rate digit                                                                   |
| <code>:Fp#</code>           | <code>0</code> or <code>1</code>              | Mode. Current implementation returns <code>1</code> for DC/pseudo-absolute, <code>0</code> otherwise. |
| <code>:FI# / :Fi#</code>    | <code>n#</code>                               | Full-in/min position                                                                                  |
| <code>:FM# / :Fm#</code>    | <code>n#</code>                               | Maximum position                                                                                      |
| <code>:Fe#</code>           | <code>n.n#</code>                             | Temperature delta from TCF baseline                                                                   |



|                     |          |                                  |
|---------------------|----------|----------------------------------|
| :Ft#                | n.n#     | Focuser temperature in C         |
| :Fu#                | n.nnnnn# | Microns per step                 |
| :FB# /<br>:Fb#      | n#       | Backlash                         |
| :FBn# /<br>:Fbn#    | 0/1      | Set backlash                     |
| :FC#                | n.nnnnn# | TCF coefficient in microns per C |
| :FCsn.n#            | 0/1      | Set TCF coefficient              |
| :Fc#                | 0 or 1   | TCF enabled status               |
| :Fc0# /<br>:Fc1#    | 0/1      | Disable/enable TCF               |
| :FD# /<br>:Fd#      | n#       | TCF deadband                     |
| :FDn# /<br>:Fdn#    | 0/1      | Set TCF deadband                 |
| :FP#                | n#       | DC motor power percent           |
| :FPn#               | 0/1      | Set DC motor power percent       |
| :FQ#                | none     | Stop focuser                     |
| :F1# ..<br>:F9#     | none     | Set move/goto rate preset        |
| :FW#                | n#       | Working goto rate in um/s        |
| :F+#                | none     | Move inward                      |
| :F-#                | none     | Move outward                     |
| :FG# /<br>:Fg#      | sn#      | Current position                 |
| :FRsn# /<br>:Frnsn# | none     | Relative goto                    |

|                     |       |                                                                |
|---------------------|-------|----------------------------------------------------------------|
| :FSn# /<br>:Fsn#    | 0/1   | Absolute goto                                                  |
| :FZ#                | none  | Zero current position                                          |
| :FH#                | none  | Set current position as home                                   |
| :Fh#                | none  | Move to home                                                   |
| :GXU4# ..<br>:GXU9# | flags | Driver status for focuser axes that expose<br>Axis.command.cpp |

Rate preset meaning:

| Preset | Meaning                   |
|--------|---------------------------|
| 1      | 1 um/s move rate          |
| 2      | 10 um/s move rate         |
| 3      | 100 um/s move rate        |
| 4      | 0.5x goto rate, move mode |
| 5      | 0.5x goto rate            |
| 6      | 0.66x goto rate           |
| 7      | 1x goto rate              |
| 8      | 1.5x goto rate            |
| 9      | 2x goto rate              |

## Rotator

Available in rotator-enabled or remote-rotator builds.

| Command | Reply | Description                                        |
|---------|-------|----------------------------------------------------|
| :rA#    | 0/1   | Rotator active                                     |
| :hP#    | 0/1   | Park rotator in standalone/remote rotator builds   |
| :hR#    | 0/1   | Unpark rotator in standalone/remote rotator builds |

|                 |                      |                                                    |
|-----------------|----------------------|----------------------------------------------------|
| :rT#            | M1# , SD3# ,<br>etc. | Status string plus rate digit                      |
| :rI#            | n#                   | Minimum angle in degrees                           |
| :rM#            | n#                   | Maximum angle in degrees                           |
| :rD#            | n.n#                 | Degrees per step                                   |
| :rb#            | n#                   | Backlash in steps                                  |
| :rbn#           | 0/1                  | Set backlash                                       |
| :rQ#            | none                 | Stop motion                                        |
| :r1# ..<br>:r9# | none                 | Set move/goto rate preset                          |
| :rW#            | n.n#                 | Working slew rate in deg/s                         |
| :rc#            | none                 | Continuous-move no-op, accepted for compatibility  |
| :r>#            | none                 | Move clockwise                                     |
| :r<#            | none                 | Move counter-clockwise                             |
| :rG#            | sDDD*MM#             | Current angle                                      |
| :rrsDDD*MM#     | none                 | Relative goto                                      |
| :rSsDDD*MM#     | 0/1                  | Absolute goto                                      |
| :rZ#            | none                 | Zero position                                      |
| :rF#            | none                 | Set current position to half travel                |
| :rC#            | none                 | Move to half-travel / home target                  |
| :r+#            | none                 | Enable derotation                                  |
| :r-#            | none                 | Disable derotation                                 |
| :rP#            | none                 | Move to parallactic angle                          |
| :rR#            | none                 | Toggle derotator reverse direction                 |
| :GX98#          | D# , R# , or         | Rotator capability: derotate, rotate-only, or none |

|        |       |                                                            |
|--------|-------|------------------------------------------------------------|
|        | N#    |                                                            |
| :GXU3# | flags | Driver status for rotator axis in standalone/remote builds |

Rotator rate presets:

| Preset | Meaning                          |
|--------|----------------------------------|
| 1      | 0.01 deg/s move rate             |
| 2      | 0.1 deg/s move rate              |
| 3      | 1.0 deg/s move rate              |
| 4      | 0.5x goto rate used as move rate |
| 5      | 0.5x goto rate                   |
| 6      | 0.66x goto rate                  |
| 7      | 1x goto rate                     |
| 8      | 1.5x goto rate                   |
| 9      | 2x goto rate                     |

## Auxiliary Features

Available when auxiliary features are enabled.

Feature slots are numbered 1..8.

## Discovery

| Command | Reply         | Description                                                 |
|---------|---------------|-------------------------------------------------------------|
| :GXY0#  | xxxxxxxx#     | Eight-character bitmap of active feature slots, 1 = present |
| :GXYn#  | name,purpose# | Slot name plus purpose code                                 |

## Slot State

| Command   | Reply                    | Description                     |
|-----------|--------------------------|---------------------------------|
| :GXXn#    | purpose-specific payload | Get current state for slot n    |
| :SXXn,Vv# | 0/1                      | Generic value set               |
| :SXXn,Zf# | 0/1                      | Dew-heater zero temperature     |
| :SXXn,Sf# | 0/1                      | Dew-heater span temperature     |
| :SXXn,Ef# | 0/1                      | Intervalometer exposure seconds |
| :SXXn,Df# | 0/1                      | Intervalometer delay seconds    |
| :SXXn,Cf# | 0/1                      | Intervalometer count            |

## :GXXn# Reply Shapes

The payload depends on the slot purpose:

| Purpose                                  | Reply shape                                            |
|------------------------------------------|--------------------------------------------------------|
| Switch / momentary switch / cover switch | value plus optional power telemetry                    |
| Analog output                            | value plus optional power telemetry                    |
| Dew heater                               | enabled,zero,span,deltaT plus optional power telemetry |
| Intervalometer                           | currentCount,exposure,delay,count                      |

When power monitoring is compiled in, the local ASCII implementation appends:

```
; <volts>,<amps>,<flags>
```

Where flags are a five-character string using P , C , U , V , T or ! .

## Axis / Motor / Driver Service Commands

Axis service commands are implemented by `Axis.command.cpp` . For the main telescope/mount build these are reachable for axis 1 and axis 2, if present. The rotator exposes axis3, if present. The focuser(s) expose any of axis 4 to axis 9, if present.

| Command        | Reply                    | Description                                     |
|----------------|--------------------------|-------------------------------------------------|
| :GXAa,p#       | value,min,max,type,name# | Get axis parameter p for axis a ( 1..9 )        |
| :GXAa,M#       | name#                    | Motor/driver name for axis a                    |
| :GXAa,0#       | n#                       | Parameter count for axis a                      |
| :GXSa#         | delta,velocity#          | Servo-only delta and velocity for axis a        |
| :GXUa#         | flags#                   | Stepper driver status for axis a                |
| :SXAC,0#       | 0/1                      | Use runtime NV axis settings                    |
| :SXAC,1#       | 0/1                      | Use compile-time Config.h axis settings         |
| :SXAa,R#       | 0/1                      | Revert axis a settings to defaults on next boot |
| :SXAa,p,value# | 0/1                      | Set axis parameter p for axis a                 |

## :GXAa,p# Reply Format

The reply is:

value,min,max,type,name#

Where:

- value is the current NV value
- min and max are the documented range
- type is the axis parameter type code returned by firmware
- name is the parameter name from the axis parameter table
- some name values are locale tokens such as \$1 or \$12 rather than literal English labels

Axis parameter type codes:

| Code | Symbol      | Meaning                     |
|------|-------------|-----------------------------|
| 0    | AXP_INVALID | Invalid / placeholder entry |

|    |                       |                                                                         |
|----|-----------------------|-------------------------------------------------------------------------|
| 1  | AXP_BOOLEAN           | Boolean value stored in NV and applied on restart/reinit                |
| 2  | AXP_BOOLEAN_IMMEDIATE | Boolean value applied immediately when set                              |
| 3  | AXP_INTEGER           | Integer value stored in NV and applied on restart/reinit                |
| 4  | AXP_INTEGER_IMMEDIATE | Integer value applied immediately when set                              |
| 5  | AXP_FLOAT             | Floating-point value stored in NV and reported as a plain numeric field |
| 6  | AXP_FLOAT_IMMEDIATE   | Floating-point value applied immediately when set                       |
| 7  | AXP_FLOAT_RAD         | Angular float stored internally in radians                              |
| 8  | AXP_FLOAT_RAD_INV     | Inverse angular float stored internally with radian/degree conversion   |
| 9  | AXP_POW2              | Power-of-two constrained integer-like value                             |
| 10 | AXP_DECAY             | Driver decay-mode selector                                              |

#### Protocol note:

- :GXAa,p# normalizes AXP\_FLOAT\_RAD and AXP\_FLOAT\_RAD\_INV to reply type code 5 and converts the value/range to degree-based units for transport.
- Axis parameter values are float-backed in firmware; values are always float even when the logical type is boolean or integer so the UI can show the appropriate controls.
- Boolean parameters: some use logical 0 / 1 , while others use the firmware constants OFF = -1 and ON = -2 ; so any UI can show 'True'/'False' or 'On'/'Off'.
- For AXP\_DECAY , the UI shows the matching decay-mode text for the numeric value 1=MIXED , 2=FAST , 3=SLOW , 4=SPREADCYCLE , 5=STEALTHCHOP .

#### Locale-backed axis parameter name tokens currently used by firmware:

| Token | Meaning         |
|-------|-----------------|
| \$1   | Steps/degree    |
| \$2   | Min limit, degs |
|       |                 |

|      |                   |
|------|-------------------|
| \$3  | Max limit, degs   |
| \$4  | Steps/um          |
| \$5  | Min limit, um     |
| \$6  | Max limit, um     |
| \$7  | Reverse           |
| \$8  | Microsteps        |
| \$9  | Microsteps Goto   |
| \$10 | Decay mode        |
| \$11 | Decay mode Goto   |
| \$12 | mA Hold           |
| \$13 | mA Run            |
| \$14 | mA Goto           |
| \$15 | 256x Interpolate  |
| \$16 | P tracking        |
| \$17 | I tracking        |
| \$18 | D tracking        |
| \$19 | P slewing         |
| \$20 | I slewing         |
| \$21 | D slewing         |
| \$22 | Rads/count        |
| \$23 | Steps/count ratio |
| \$24 | Max accel, %/s/s  |
| \$25 | Min power, %      |
| \$26 | Max power, %      |



# CAN Remote-Node Variants

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For architecture, configuration, and current scope, see [CAN\\_NOTES.md](#).

The following files implement the same logical command families over packed CAN messages rather than the ASCII LX200-style transport:

- `src/telescope/focuser/local/Focuser.can.command.cpp`
- `src/telescope/rotator/local/Rotator.can.command.cpp`
- `src/telescope/auxiliary/local/Features.can.command.cpp`

These are not separate end-user command names. They are binary transport implementations of the same focuser, rotator, and auxiliary-feature command sets above.

Notable transport differences:

- CAN replies are binary-packed, not ASCII text.
- `:GXY0#` for CAN features returns a single bitmask byte rather than an eight-character ASCII bitmap.
- CAN focuser/rotator driver-status replies are packed bitfields rather than comma-separated text flags.

## Source Coverage

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This document was built from the current handlers in:

- `src/telescope/Telescope.command.cpp`
- `src/telescope/mount/Mount.command.cpp`
- `src/telescope/mount/goto/Goto.command.cpp`
- `src/telescope/mount/guide/Guide.command.cpp`
- `src/telescope/mount/home/Home.command.cpp`
- `src/telescope/mount/library/Library.command.cpp`
- `src/telescope/mount/limits/Limits.command.cpp`
- `src/telescope/mount/park/Park.command.cpp`
- `src/telescope/mount/pec/Pec.command.cpp`
- `src/telescope/mount/site/Site.command.cpp`
- `src/telescope/mount/status/Status.command.cpp`

- `src/lib/axis/Axis.command.cpp`
- `src/telescope/focuser/local/Focuser.command.cpp`
- `src/telescope/rotator/local/Rotator.command.cpp`
- `src/telescope/auxiliary/local/Features.command.cpp`
- plus the corresponding CAN transport handlers
- plus `src/libApp/commands/ProcessCmds.cpp`